

# Thue–Morse Mellin Renormalization

Fabius-controlled derivative asymptotics,  
positive dyadic solutions, and gamma-tower uniqueness

A research extension to the ProveIt Thue–Morse atlas

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## Abstract

The signed Thue–Morse sequence has a positive Laplace product  $Q(t) = \prod_{j \geq 0} (1 - e^{-2^j t})$ . We pair it with the Fabius uniform-sum Laplace transform  $H(z) = \prod_{j \geq 1} (1 - e^{-z/2^j})/(z/2^j)$ . Their product satisfies the exact equation  $\mathcal{W}(2t) = \mathcal{W}(t)/t$ . This elementary cancellation gives a positive lognormal-type Mellin kernel and an all-orders expansion for the derivatives of the entire shifted Thue–Morse Dirichlet function at its negative-integer zeros. The coefficients are finite expressions in Bernoulli numbers and the shift parameter. A global Taylor estimate yields a uniform dyadic  $q$ -Gevrey remainder and a Gaussian-in-order bound after optimal truncation.

We also classify every completely monotone solution of the basic dyadic equation by a positive measure on one multiplicative period. This produces many positive gamma towers with the same normalization. More strongly, distinct towers can have identical formal high-order expansions to every order; their discrepancy is beyond all those corrections. Nevertheless, the single boundary condition  $-af'(a) \rightarrow 1$  as  $a \downarrow 0$  selects the canonical solution and its full differential tower. Complete proofs, explicit coefficients, positive Hankel consequences, and independently evaluated numerical checks are provided. Classical ingredients are separated from the results derived here; historical priority of the new formulations is not asserted as established.

**Keywords.** Thue–Morse sequence; Fabius function; entire Dirichlet series; Mellin transform; completely monotone function; moment problem; log-periodicity; beyond-all-orders asymptotics.

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# 1 The questions and the main answers

Let  $s_2(n)$  denote the number of ones in the binary expansion of  $n$ , and put

$$\varepsilon_n = (-1)^{s_2(n)}, \quad n = 0, 1, 2, \dots \quad (1.1)$$

The first terms are  $1, -1, -1, 1, -1, 1, 1, -1, \dots$ . Rather than attempting another catalogue of word-combinatorial properties, we investigate two analytic questions that connect directly with the Fabius–Rvachev program: how rapidly do the derivatives of the associated entire Dirichlet function grow at negative integers, and what actually characterizes its gamma-type hierarchy?

Throughout this article  $a > 0$  is a real shift,  $\lambda = \log 2$ , and

$$D(s, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n + a)^s} \quad (\operatorname{Re} s > 1) \quad (1.2)$$

denotes the Dirichlet series and subsequently its entire continuation. There are two different limiting parameters:  $\rho \rightarrow \infty$  describes *high negative order*, whereas  $a \downarrow 0$  describes a *boundary of the shift variable*. They should not be conflated.

## 1.1 High negative orders are governed by a positive dyadic kernel

Define

$$\begin{aligned} Q(t) &= \prod_{j \geq 0} (1 - e^{-2^j t}), & H(z) &= \prod_{j \geq 1} \frac{1 - e^{-z/2^j}}{z/2^j}, \\ \mathcal{W}(t) &= Q(t)H(t), & C_\rho(a) &= \int_0^\infty t^{-\rho-1} e^{-at} Q(t) dt. \end{aligned} \quad (1.3)$$

Each removable value in the product for  $H$  is understood by continuity. For an integer  $m \geq 0$ ,

$$\partial_s D(-m, a) = (-1)^m m! C_m(a). \quad (1.4)$$

The important cancellation is

$$\mathcal{W}(2t) = \frac{\mathcal{W}(t)}{t}. \quad (1.5)$$

It yields a positive one-periodic amplitude  $\Theta$  and coefficient polynomials  $c_k(a)$  such that

$$\frac{C_\rho(a)}{2^{\rho(\rho-1)/2} \Theta(\rho)} \sim \sum_{k \geq 0} c_k(a) 2^{-k\rho}. \quad (1.6)$$

The expansion is accompanied by an explicit-form remainder bound, not merely formal coefficient matching. At integer orders  $\Theta(m) = K_0$ , where

$$K_0 = \int_0^\infty \mathcal{W}(t) \frac{dt}{t} = 0.75072198040787095920970791405063469 \dots \quad (1.7)$$

For example, at the symmetric shift  $a = 1/2$ ,

$$\frac{C_m(1/2)}{K_0 2^{m(m-1)/2}} \sim 1 - \frac{2^{-2m}}{9} + \frac{248}{2025} 2^{-4m} - \frac{4806656}{2679075} 2^{-6m} + \dots \quad (1.8)$$

All odd corrections vanish. Section 4 proves these statements, including uniformity and optimal truncation.

## 1.2 The dyadic equation has a large positive solution space

A function  $f : (0, \infty) \rightarrow \mathbb{R}$  is *completely monotone* if it is smooth and  $(-1)^j f^{(j)}(a) \geq 0$  for every  $j \geq 0$ . We completely describe the functions in this class satisfying

$$f(a) = f(a/2) - f((a+1)/2). \quad (1.9)$$

They correspond exactly to positive, locally finite measures  $\nu$  on  $(0, \infty)$  invariant under multiplication by two:

$$f(a) = \int_0^\infty e^{-at} Q(t) \nu(dt), \quad \nu(2B) = \nu(B). \quad (1.10)$$

The canonical solution  $C_0(a)$  corresponds to  $\nu(dt) = dt/t$ . The normalization  $f(1/2) = \log 2$  fixes only the mass of one multiplicative period, not the distribution of that mass. Even the addition of all formal high-order coefficients need not restore uniqueness (Section 7).

## 1.3 A boundary condition does restore uniqueness

Within the completely monotone class, the equation (1.9) together with

$$\lim_{a \downarrow 0} (-a f'(a)) = 1 \quad (1.11)$$

characterizes  $f = C_0$  uniquely. The proof identifies the representing measure from all dyadic boundary limits. The stronger, convenient condition that  $f(a) + \log a$  have a finite limit also suffices. Together with the differential ladder, this selects the entire gamma tower. These are precise characterization statements; they do not claim that an arbitrary gamma-like function is uniquely determined without a specified function class or boundary condition.

## 1.4 Relation to the atlas and the literature

The supplied atlas [1] develops finite products, repeated summation, Fabius–Rvachev connections, an entire Mellin continuation, and an automatic gamma hierarchy. Its “Automatic Barnes hierarchy” frontier asks for normalizations yielding a uniqueness theorem. The present work gives a complete answer within the completely monotone class, and also exhibits a specific obstruction to using high-order asymptotics alone.

The analytic study of Thue–Morse Dirichlet series predates this article by decades; relevant works include Allouche–Cohen [2], Allouche–Mendès France–Peyrière [3], and Allouche’s later survey [4]. Explicit linear-combination identities give a complementary direction [9]. In particular, the simplicity of the negative-integer zeros is *not* offered as a new result. The Fabius probability and product background is also classical; see Arias de Reyna [5]. Log-periodic effects in Mellin analysis and moment indeterminacy of lognormal-type laws have established precedents [6, 7].

The contributions developed and proved here are the quantitative renormalization theorem with its global remainder, the measure classification and boundary characterization, and the construction of distinct normalized towers sharing every formal high-order correction. The exact product pairing and its subsidiary consequences organize their proofs. These are candidate research contributions, not a claim of verified historical priority. The targeted comparison did not locate the same main statements in the inspected atlas or accessible literature, but it was not an exhaustive search. In particular, the author’s link to Alkauskas’s 2001 preprint [10] could not be retrieved; no claim about its detailed contents is made. This manuscript was prepared with ChatGPT; it has not been peer-reviewed or formalized in Lean.

## 2 From binary signs to an entire Mellin transform

### 2.1 The product and elementary endpoint bounds

The two binary identities

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n \quad (2.1)$$

show by induction that

$$\sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (2.2)$$

For  $|z| < 1$ , both sides converge as  $m \rightarrow \infty$ , so

$$Q(t) = \sum_{n \geq 0} \varepsilon_n e^{-nt} \quad (t > 0). \quad (2.3)$$

Although the sum has signs, the product is strictly positive.

**Lemma 2.1** (Bounds needed for all Mellin operations). *The product  $Q$  is positive and smooth on  $(0, \infty)$  and satisfies*

$$Q(t) = (1 - e^{-t})Q(2t). \quad (2.4)$$

For every integer  $N \geq 0$ ,

$$0 < Q(t) \leq 2^{N(N-1)/2} t^N \quad (t > 0), \quad (2.5)$$

and

$$0 \leq 1 - Q(t) \leq \frac{e^{-t}}{1 - e^{-t}}. \quad (2.6)$$

In particular,  $Q$  is smaller than any fixed power at zero and approaches one exponentially at infinity.

*Proof.* Uniform convergence of the product and its differentiated logarithms on compact subintervals of  $(0, \infty)$  follows from the exponential growth of  $2^j$ . Its factors are positive, and the sum of their deficits is finite. The scaling equation follows by shifting the index. For (2.5), bound the first  $N$  factors by  $2^j t$ , and the rest by one. For (2.6), use  $1 - \prod_j (1 - u_j) \leq \sum_j u_j$  for  $0 \leq u_j \leq 1$ , then  $2^j \geq j + 1$ .  $\square$

### 2.2 Continuation, real zeros, and positive derivative data

**Proposition 2.2** (Entire continuation and real zeros). *For  $a > 0$ , the function*

$$M(s, a) = \int_0^\infty t^{s-1} e^{-at} Q(t) dt, \quad D(s, a) = \frac{M(s, a)}{\Gamma(s)}, \quad (2.7)$$

*is entire in  $s$  and agrees with (1.2) where  $\operatorname{Re} s > 1$ . Its real zeros are exactly  $0, -1, -2, \dots$ , and every one is simple. Moreover, (1.4) holds.*

*Proof.* Fix a compact set of  $s$ -values. At zero, choose  $N$  in (2.5) large enough to dominate  $t^{\operatorname{Re} s - 1} |\log t|^j Q(t)$  for every derivative order  $j$  under consideration. At infinity,  $e^{-at}$  dominates any such power and logarithm. Differentiation under the integral is therefore locally uniform, which proves that  $M$  is entire.

For  $\operatorname{Re} s > 1$ , termwise integration of (2.3) is justified by the corresponding absolutely convergent unsigned sum. It gives the usual gamma integral for each  $(n + a)^{-s}$ . For real  $s$ , the integral  $M(s, a)$  is strictly positive. The reciprocal gamma function has simple zeros precisely at the nonpositive integers, with

$$\left. \frac{d}{ds} \frac{1}{\Gamma(s)} \right|_{s=-m} = (-1)^m m!.$$

Multiplication by  $M(s, a)$  proves every assertion.  $\square$

It is useful to keep  $\rho$  real rather than immediately restrict it to an integer. The positive quantities  $C_\rho$  in (1.3) exist for every real  $\rho$ , although the asymptotic statements below will use  $\rho \geq 0$ . For every integer  $\ell \geq 0$ ,

$$(-1)^\ell \partial_a^\ell C_\rho(a) = \int_0^\infty t^{\ell-\rho-1} e^{-at} Q(t) dt > 0. \quad (2.8)$$

Thus every  $C_\rho$  is completely monotone. In particular,

$$C'_m(a) = -C_{m-1}(a) \quad (m \geq 1). \quad (2.9)$$

Using (2.4) and substituting  $t = 2u$  gives

$$C_\rho(a) = 2^\rho \{C_\rho(a/2) - C_\rho((a+1)/2)\}. \quad (2.10)$$

All of these identities follow from absolutely convergent integrals; regularized sums are unnecessary here.

### 2.3 The Fabius uniform-sum transform

Let  $U_1, U_2, \dots$  be independent uniform random variables on  $[0, 1]$ , and set

$$X = \sum_{j \geq 1} 2^{-j} U_j. \quad (2.11)$$

This converges almost surely, takes values in  $[0, 1]$ , and has mean  $1/2$ . The distribution is the standard dyadic uniform-sum realization underlying the Fabius function. Explicitly fixing (2.11) avoids a possible factor-of-two ambiguity among Fabius and Rvachev conventions. Independence gives

$$H(z) = \mathbb{E} e^{-zX} = \prod_{j \geq 1} \psi(z/2^j), \quad \psi(z) = \frac{1 - e^{-z}}{z}, \quad \psi(0) = 1. \quad (2.12)$$

Boundedness of  $X$  proves that  $H$  is entire. Equivalently, the product converges locally uniformly since  $\psi(z/2^j) = 1 + O_K(2^{-j})$  on every compact set  $K$ .

The distributional identity  $X \stackrel{d}{=} (U + X')/2$ , with  $X'$  an independent copy, gives a direct connection with the usual functional equation. If  $F(x) = \Pr(X \leq x)$  is extended by zero to  $x \leq 0$  and by one to  $x \geq 1$ , convolution with the uniform law gives

$$F'(x) = 2\{F(2x) - F(2x - 1)\}.$$

Consequently  $F'(x) = 2F(2x)$  on  $0 < x < 1/2$ . For every  $N \geq 2$ , the sum of the first  $N$  uniform terms has a  $C^{N-2}$  spline density; convolution with the remaining probability law preserves that regularity. Since  $N$  is arbitrary, the density is smooth. Only the entire transform (2.12) is needed below.

**Lemma 2.3** (Scaling, symmetry, and singularity distance). *The function  $H$  satisfies*

$$H(2z) = \psi(z)H(z), \quad H(-z) = e^z H(z). \quad (2.13)$$

*Its zeros are exactly the nonzero integer multiples of  $4\pi i$ , with multiplicities prescribed by the individual factors. The nearest zeros to zero,  $\pm 4\pi i$ , are simple. In  $|z| < 4\pi$ ,*

$$\log H(z) = -\frac{z}{2} + \sum_{j \geq 1} b_j z^{2j}, \quad b_j = \frac{B_{2j}}{2j(2j)!(4^j - 1)}. \quad (2.14)$$

*Here the logarithm has value zero at the origin.*

*Proof.* The first identity is an index shift. The second follows either from  $X \stackrel{d}{=} 1 - X$  or directly from  $\psi(-z) = e^z \psi(z)$ . Each factor has zeros at  $z = 2^{j+1} \pi i k$ ,  $k \neq 0$ . Away from those zeros, the convergence of the logarithmic tail shows that the product is nonzero. This proves the description of all zeros and their nearest distance.

The Bernoulli expansion is obtained by integrating the power series for  $\frac{1}{2} \coth(z/2) - 1/z$ :

$$\log \psi(z) = -\frac{z}{2} + \sum_{j \geq 1} \frac{B_{2j}}{2j(2j)!} z^{2j}, \quad |z| < 2\pi.$$

Substitute  $z/2^\ell$  and sum the resulting geometric series over  $\ell \geq 1$ . The nearest factor singularity moves the resulting radius to  $4\pi$ , yielding (2.14).  $\square$

### 3 An exact lognormal-type kernel

#### 3.1 The product cancellation and its envelope

**Theorem 3.1** (Exact Mellin renormalization kernel). *Let  $\mathcal{W}(t) = Q(t)H(t)$ . Then  $\mathcal{W} > 0$  and*

$$\mathcal{W}(2t) = t^{-1} \mathcal{W}(t), \quad \mathcal{W}(2^j v) = 2^{-j(j-1)/2} v^{-j} \mathcal{W}(v) \quad (j \in \mathbb{Z}). \quad (3.1)$$

*The function*

$$A(y) = 2^{(y^2-y)/2} \mathcal{W}(2^y) \quad (3.2)$$

*is positive, smooth, and one-periodic. Hence*

$$\mathcal{W}(e^x) = \exp\left(-\frac{x^2}{2\lambda} + \frac{x}{2}\right) A(x/\lambda). \quad (3.3)$$

*There are constants  $0 < A_- \leq A_+ < \infty$  such that the expression on the right with  $A$  replaced by  $A_-$  or  $A_+$  is respectively a lower or an upper bound for  $\mathcal{W}(e^x)$ , for every real  $x$ .*

*Proof.* From (2.4) and (2.13),

$$\mathcal{W}(2t) = \frac{Q(t)}{1 - e^{-t}} \frac{1 - e^{-t}}{t} H(t) = \frac{\mathcal{W}(t)}{t}.$$

Iteration gives the second identity for positive integers  $j$ , and solving backwards gives it for negative integers as well. Substitution into (3.2) gives  $A(y+1) = A(y)$ . Positivity and smoothness follow from those of the two factors. The positive minimum and finite maximum of  $A$  on  $[0, 1]$  give the bounds.  $\square$

The word “lognormal-type” refers to the Gaussian in  $x = \log t$ . It does *not* assert that  $\mathcal{W}(t) dt/t$  is itself exactly a lognormal law: the periodic factor  $A$  remains present.

### 3.2 The Mellin recurrence and phase amplitude

Define

$$K(s) = \int_0^\infty t^{s-1} \mathcal{W}(t) dt \quad (s \in \mathbb{C}). \quad (3.4)$$

The two-sided Gaussian envelope proves local uniform absolute convergence for every  $s$ , including every differentiated integral. Thus  $K$  is entire and is strictly positive on the real axis.

**Proposition 3.2** (Exact moments and a periodic amplitude). *For  $s \in \mathbb{C}$  and  $k \in \mathbb{Z}$ ,*

$$K(s+1) = 2^{s+1} K(s), \quad K(s+k) = 2^{ks+k(k+1)/2} K(s). \quad (3.5)$$

*In particular,  $K(k) = 2^{k(k+1)/2} K_0$  for every integer  $k$ . The function*

$$\Theta(\rho) = 2^{-\rho(\rho-1)/2} K(-\rho) \quad (3.6)$$

*is entire, one-periodic, and positive for real  $\rho$ .*

*Proof.* Substitute  $t = 2u$  in (3.4) and use (3.1):  $K(s) = 2^s K(s-1)$ . This yields the first recurrence and then its integer iterates. Substituting the recurrence in (3.6) proves periodicity. The other statements have already been justified by the integral.  $\square$

### 3.3 Why the Mellin modulation is extremely small

The periodic function  $A$  has an absolutely convergent Fourier series

$$A(y) = \sum_{\ell \in \mathbb{Z}} a_\ell e^{2\pi i \ell y}, \quad a_\ell = \int_0^1 A(y) e^{-2\pi i \ell y} dy. \quad (3.7)$$

For example, two integrations by parts already give a summable bound on its Fourier coefficients. Integrating each Fourier mode against the Gaussian in (3.3) gives

$$K(s) = \sqrt{2\pi\lambda} e^{\lambda(s+1/2)^2/2} \sum_{\ell \in \mathbb{Z}} a_\ell e^{-2\pi^2 \ell^2 / \lambda} e^{2\pi i \ell (s+1/2)}. \quad (3.8)$$

The identity holds initially for real  $s$ , where absolute convergence justifies all operations, and then for complex  $s$  by analytic continuation. Consequently,

$$\Theta(\rho) = \sqrt{2\pi\lambda} 2^{1/8} \sum_{\ell \in \mathbb{Z}} (-1)^\ell a_\ell e^{-2\pi^2 \ell^2 / \lambda} e^{-2\pi i \ell \rho}. \quad (3.9)$$

Mellin transformation suppresses the  $\ell$ th oscillatory mode by a Gaussian factor in  $\ell$ . Section 9 finds relative variations of  $\Theta$  around  $10^{-18}$ . These numerical observations are not used as a proof that  $\Theta$  is nonconstant; none of our main theorems requires that additional assertion.

### 3.4 Two exact expansions from the same periodic factor

The identity  $Q = \mathcal{W}/H$ , together with (2.14), gives the convergent small- $t$  correction formula

$$\begin{aligned} \log Q(t) = & -\frac{(\log t)^2}{2\lambda} + \frac{1}{2} \log t + \log A(\log_2 t) \\ & + \frac{t}{2} - \sum_{j \geq 1} b_j t^{2j}, \quad 0 < t < 4\pi. \end{aligned} \quad (3.10)$$



At the other end, expanding  $-\log Q(t)$  in exponentials gives

$$\begin{aligned} \log H(t) = & -\frac{(\log t)^2}{2\lambda} + \frac{1}{2}\log t + \log A(\log_2 t) \\ & + \sum_{n \geq 1} \frac{2^{v_2(n)+1} - 1}{n} e^{-nt}, \quad t > 0, \end{aligned} \quad (3.11)$$

where  $v_2(n)$  is the exponent of two dividing  $n$ . Indeed, in  $\sum_{j \geq 0} \sum_{r \geq 1} e^{-r2^j t}/r$ , the coefficient of  $e^{-nt}$  is  $n^{-1} \sum_{j=0}^{v_2(n)} 2^j$ . All terms are nonnegative, so the rearrangement is justified by Tonelli's theorem. Thus the same periodic amplitude links a Bernoulli correction at one endpoint to a convergent exponential correction at the other.

## 4 All-orders derivative growth with a uniform remainder

### 4.1 Coefficient polynomials

Define

$$R_a(z) = \frac{e^{-az}}{H(z)} = \sum_{k \geq 0} r_k(a) z^k, \quad c_k(a) = 2^{k(k+1)/2} r_k(a). \quad (4.1)$$

The Taylor series has radius exactly  $4\pi$ , since the numerator cannot cancel the simple poles at  $\pm 4\pi i$ . Put

$$L_1 = \frac{1}{2} - a, \quad L_{2j} = -b_j \ (j \geq 1), \quad L_{2j+1} = 0 \ (j \geq 1). \quad (4.2)$$

Then  $R_a(z) = \exp(\sum_{\ell \geq 1} L_\ell z^\ell)$  near zero, and a finite, nonrecursive coefficient formula is

$$c_k(a) = 2^{k(k+1)/2} \sum_{\substack{j_1, \dots, j_k \geq 0 \\ j_1 + 2j_2 + \dots + kj_k = k}} \prod_{\ell=1}^k \frac{L_\ell^{j_\ell}}{j_\ell!}, \quad c_0 = 1. \quad (4.3)$$

For computation, the equivalent recurrence

$$nr_n = \sum_{\ell=1}^n \ell L_\ell r_{n-\ell}, \quad r_0 = 1, \quad (4.4)$$

is often more economical. It is a convenience, not a prerequisite to the closed expression (4.3).

With  $b = a - 1/2$ , the first coefficients are

$$\begin{aligned} c_0 &= 1, & c_1 &= -2b, \\ c_2 &= 4b^2 - \frac{1}{9}, & c_3 &= -\frac{32}{3}b^3 + \frac{8}{9}b, \\ c_4 &= \frac{8}{2025}(10800b^4 - 1800b^2 + 31), \\ c_5 &= -\frac{256b}{2025}(2160b^4 - 600b^2 + 31), \\ c_6 &= \frac{2048}{2679075}(3810240b^6 - 1587600b^4 + 164052b^2 - 2347). \end{aligned} \quad (4.5)$$

Differentiation and reflection give useful independent checks:

$$c'_k(a) = -2^k c_{k-1}(a), \quad c_k(1-a) = (-1)^k c_k(a). \quad (4.6)$$

For the second identity, use  $R_{1-a}(z) = R_a(-z)$ .

## 4.2 Statement of the uniform theorem

For a compact interval  $I \subset (0, \infty)$  and  $0 < r < 4\pi$ , set

$$\mathcal{T}_r = \{z \in \mathbb{C} : \text{dist}(z, [0, \infty)) \leq r\}, \quad M_{I,r} = \sup_{a \in I, z \in \mathcal{T}_r} |R_a(z)|. \quad (4.7)$$

The finiteness of this constant is part of the proof below. It is a mathematically specified bound; no numerical enclosure for  $M_{I,r}$  is claimed in the computational tables.

**Theorem 4.1** (Uniform all-orders expansion). *For every compact  $I \subset (0, \infty)$ , every  $0 < r < 4\pi$ , every integer  $N \geq 1$ , every  $a \in I$ , and every real  $\rho \geq 0$ ,*

$$\frac{C_\rho(a)}{2^{\rho(\rho-1)/2}\Theta(\rho)} = \sum_{k=0}^{N-1} c_k(a)2^{-k\rho} + \mathcal{R}_N(\rho, a), \quad (4.8)$$

where

$$|\mathcal{R}_N(\rho, a)| \leq M_{I,r} r^{-N} 2^{N(N+1)/2 - N\rho}. \quad (4.9)$$

In particular, for each fixed  $N$  the error is uniformly  $O_{I,N}(2^{-N\rho})$  as  $\rho \rightarrow \infty$ .

The estimate is useful both with fixed  $N$  and with  $N$  increasing with  $\rho$ . The latter use is the reason for keeping the dependence on  $N$  explicit.

## 4.3 Step one: exact finite-level renormalization

**Lemma 4.2.** *Write  $\rho = m + \theta$ , where  $m = \lfloor \rho \rfloor$  and  $0 \leq \theta < 1$ , and set  $h = 2^{-m}$ . Then*

$$C_{m+\theta}(a) = 2^{m(m-1)/2+m\theta} \int_0^\infty t^{-\theta-1} \mathcal{W}(t) R_a(ht) dt. \quad (4.10)$$

*Proof.* Iteration of the product equation gives

$$Q(2^{-m}t) = Q(t) \prod_{j=1}^m (1 - e^{-t/2^j}).$$

Separate each factor into  $(t/2^j)\psi(t/2^j)$  and use the product for  $H$ :

$$Q(2^{-m}t) = 2^{-m(m+1)/2} t^m Q(t) \frac{H(t)}{H(2^{-m}t)}. \quad (4.11)$$

Substituting the old variable  $2^{-m}t$  in the definition of  $C_{m+\theta}$  now gives (4.10), with all powers of two displayed.  $\square$

The point of this formula is that the rapidly changing order  $m$  has been removed from the Mellin weight. It occurs only in the explicit prefactor and in the small argument of  $R_a$ .

#### 4.4 Step two: a Taylor bound valid on the entire positive axis

**Lemma 4.3** (Boundedness on a pole-free tube). *For  $I \subset (0, \infty)$  compact and  $r < 4\pi$ , the constant  $M_{I,r}$  in (4.7) is finite. Consequently, for every real  $x \geq 0$ ,*

$$\left| R_a(x) - \sum_{k=0}^{N-1} r_k(a) x^k \right| \leq M_{I,r} r^{-N} x^N \quad (a \in I). \quad (4.12)$$

*Proof.* The tube contains no zero of  $H$ . Thus  $R_a$  is jointly continuous and uniformly bounded when  $a \in I$  and  $z$  stays in a compact portion of the tube. It remains to control its unbounded right-hand end.

For real  $x > 0$ , let

$$\mu_x = \frac{\mathbb{E}[X e^{-xX}]}{\mathbb{E}[e^{-xX}]} = -\frac{H'(x)}{H(x)}.$$

Logarithmic differentiation of the independent factors gives

$$\mu_x = \sum_{j \geq 1} \left( \frac{1}{x} - \frac{2^{-j}}{e^{x2^{-j}} - 1} \right). \quad (4.13)$$

Each summand is the mean of a uniform variable on  $[0, 2^{-j}]$  under a positive exponential tilt; it lies between zero and  $2^{-j}/2$ , and is also at most  $1/x$ . The upper bound by the untilted mean follows because the derivative of a tilted mean is the negative tilted variance. Splitting at  $J = \lfloor \log_2 x \rfloor$  for  $x \geq 1$  therefore gives

$$0 \leq \mu_x \leq \frac{\log_2 x + 2}{x}. \quad (4.14)$$

For  $|y| \leq r$ ,

$$|H(x + iy) - H(x)| \leq \mathbb{E}[e^{-xX} |e^{-iyX} - 1|] \leq |y| \mu_x H(x).$$

The rightmost multiplier is at most  $1/2$  for sufficiently large  $x$ , uniformly in  $|y| \leq r$ . Thus  $|H(x + iy)| \geq H(x)/2$  there.

Since  $Q(x) \leq 1$ , we have  $H(x) = \mathcal{W}(x)/Q(x) \geq \mathcal{W}(x)$ . The lower bound in Theorem 3.1, with  $a_0 = \min I > 0$ , yields

$$|R_a(x + iy)| \leq \frac{2}{A_-} \exp \left[ -a_0 x + \frac{(\log x)^2}{2\lambda} - \frac{1}{2} \log x \right] \rightarrow 0.$$

The remaining part of the tube with negative real part is compact. This proves  $M_{I,r} < \infty$ .

Every closed disc of radius  $r$  centered on  $[0, \infty)$  lies in the tube. Cauchy's estimate gives  $|R_a^{(N)}(x)|/N! \leq M_{I,r} r^{-N}$  for all  $x \geq 0$ . Taylor's integral remainder along the real interval from 0 to  $x$  gives (4.12). Notice that this argument does not require  $x < 4\pi$ ; the radius restriction is on the complex discs used to control derivatives, not on the real argument of the final estimate.  $\square$

#### 4.5 Step three: integration of the global remainder

*Proof of Theorem 4.1.* Insert the finite Taylor polynomial from (4.12) into (4.10). The  $k$ th term integrates to  $r_k(a) h^k K(k - \theta)$ . The shift identity gives

$$\frac{K(k - \theta)}{K(-\theta)} = 2^{-k\theta + k(k+1)/2}. \quad (4.15)$$

Also,

$$2^{m(m-1)/2+m\theta} K(-\theta) = 2^{\rho(\rho-1)/2} \Theta(\rho). \quad (4.16)$$

Dividing the finite sum by this leading factor yields  $c_k(a)2^{-k\rho}$  term by term.

Positivity of the weight  $t^{-\theta-1}\mathcal{W}(t)$  permits direct integration of the absolute remainder bound. The resulting relative error is at most

$$M_{I,r} r^{-N} h^N \frac{K(N-\theta)}{K(-\theta)} = M_{I,r} r^{-N} 2^{N(N+1)/2-N\rho}.$$

This proves the estimate without interchanging an infinite Taylor series and an improper integral.  $\square$

#### 4.6 Divergence, optimal truncation, and centering

The Taylor series for  $R_a$  has radius  $4\pi$ , so

$$\limsup_{k \rightarrow \infty} |r_k(a)|^{1/k} = \frac{1}{4\pi}. \quad (4.17)$$

Along an infinite subsequence the additional factor  $2^{k(k+1)/2}$  forces  $|c_k(a)|^{1/k} \rightarrow \infty$ . Hence the formal power series  $\sum c_k(a)h^k$  has radius zero for every  $a > 0$ . Its use in (4.8) is asymptotic, not convergent. For any  $r < 4\pi$ , Cauchy's estimate also gives  $|c_k(a)| \leq M_{I,r} r^{-k} 2^{k(k+1)/2}$ . We call this specific bound *dyadic  $q$ -Gevrey growth*; the displayed formula fixes the convention, rather than relying on differing definitions of  $q$ -Gevrey order in the literature.

Let

$$\tau = \rho + \log_2 r - \frac{1}{2}. \quad (4.18)$$

The base-two logarithm of the order-dependent factor in (4.9) is  $\frac{1}{2}(N-\tau)^2 - \frac{1}{2}\tau^2$ . If  $\tau \geq 1$ , choose a nearest integer  $N \geq 1$ . Then

$$|\mathcal{R}_N(\rho, a)| \leq 2^{1/8} M_{I,r} 2^{-\frac{1}{2}(\rho + \log_2 r - 1/2)^2}. \quad (4.19)$$

This is an achievable upper bound after a specified truncation, not a claim that the true error has exactly this size or that the estimate is asymptotically sharp.

When  $a = 1/2$ , the function  $R_{1/2}$  is even. Therefore all odd coefficients vanish and the small scale in (1.8) is  $4^{-m}$  instead of  $2^{-m}$ . Fixed-order errors may accordingly be improved by retaining the next vanishing odd coefficient before applying the theorem. For example, the error after the displayed  $2^{-6m}$  term is  $O(2^{-8m})$ .

#### 4.7 Growth of the entire function and its gamma tower

Define the gamma hierarchy by

$$\Gamma_m^{\text{TM}}(a) = \exp(\partial_s D(-m, a)) = \exp((-1)^m m! C_m(a)). \quad (4.20)$$

Thus the expansion describes the *signed logarithms* of these functions. Exponentiating a truncated logarithmic expansion without tracking the very large factor  $m!2^{m(m-1)/2}$  would not automatically produce a small relative error for  $\Gamma_m^{\text{TM}}$  itself. This distinction matters at high order.

**Corollary 4.4** (Order and type). *For every fixed  $a > 0$ , the entire function  $s \mapsto D(s, a)$  has order two and type  $\lambda/2$ . Equivalently, if  $M_D(R) = \max_{|s| \leq R} |D(s, a)|$ , then*

$$\limsup_{R \rightarrow \infty} \frac{\log M_D(R)}{R^2} = \frac{\log 2}{2}. \quad (4.21)$$

*Proof.* For  $R \geq 2$  and  $|s| \leq R$ , split the Mellin integral at one:

$$|M(s, a)| \leq C_R(a) + \Gamma(R)a^{-R}.$$

Theorem 4.1, periodic boundedness of  $\Theta$ , and the usual gamma integral estimate give a logarithmic upper bound  $\lambda R^2/2 + O_a(R \log R)$ . The Weierstrass product for  $1/\Gamma(s)$  gives  $\max_{|s| \leq R} |1/\Gamma(s)| \leq \exp(O(R \log R))$ . For completeness, split its product at  $n = \lceil 2R \rceil$ : the first part contributes  $O(R \log R)$  and  $\log(1 + s/n) - s/n = O(R^2/n^2)$  controls the tail. This proves the required upper bound for  $D$ .

For the reverse bound, Cauchy’s estimate on the circle of radius one about  $-m$  gives

$$M_D(m+1) \geq |\partial_s D(-m, a)| = m! C_m(a).$$

The main theorem implies  $\log C_m(a) = \lambda m(m-1)/2 + \log K_0 + o(1)$ . Dividing by  $(m+1)^2$  proves the matching lower limit superior. A positive finite limit superior in (4.21) establishes the stated order and type.  $\square$

## 5 Positive moments, Hankel determinants, and invisible phases

### 5.1 Strict positivity and quantitative Hankel growth

The substitution  $x = 1/t$  in  $C_m$  gives

$$C_m(a) = \int_0^\infty x^m w_a(x) dx, \quad w_a(x) = x^{-1} e^{-a/x} Q(1/x) > 0. \quad (5.1)$$

Every integer moment exists. Thus the following positivity concerns the *derivative data* of  $D$ , not an unrelated Hankel matrix formed straight from the signs  $\varepsilon_n$ .

**Proposition 5.1** (Strict Hankel positivity). *For integers  $n, \ell \geq 0$ ,*

$$\det[C_{i+j+\ell}(a)]_{i,j=0}^n > 0. \quad (5.2)$$

*In particular,*

$$C_m(a)C_{m+2}(a) > C_{m+1}(a)^2, \quad \frac{C_m(a)C_{m+2}(a)}{C_{m+1}(a)^2} \rightarrow 2. \quad (5.3)$$

*More generally, for fixed  $n$  as  $m \rightarrow \infty$ ,*

$$\begin{aligned} \det[C_{m+i+j}(a)]_{i,j=0}^n &\sim K_0^{n+1} 2^{\frac{n+1}{2}m(m-1)+mn(n+1)+\frac{n(n+1)(n-1)}{3}} \\ &\times \prod_{0 \leq i < j \leq n} (2^j - 2^i). \end{aligned} \quad (5.4)$$

*The relative error in (5.4) is  $O(2^{-m})$  for fixed  $a, n$ , and is  $O(2^{-2m})$  at  $a = 1/2$ .*

*Proof.* The matrix in (5.2) is the Gram matrix of  $1, x, \dots, x^n$  for the strictly positive measure  $x^\ell w_a(x) dx$ . A nonzero polynomial cannot vanish almost everywhere for this measure, so the matrix is positive definite.

For the asymptotic determinant, apply Theorem 4.1 to its finitely many entries. Factor  $K_0 2^{m(m-1)/2}$  from each row,  $2^{mi+i(i-1)/2}$  from row  $i$ , and  $2^{mj+j(j-1)/2}$  from column  $j$ . The remaining matrix is  $[2^{ij} + O(2^{-m})]$ . Its limiting determinant is the nonzero Vandermonde product  $\prod_{i < j} (2^j - 2^i)$ . Summing the row and column exponents gives (5.4); the centered improvement follows from  $c_1(1/2) = 0$ . The limiting ratio in (5.3) follows directly from the leading terms of the three entries.  $\square$

## 5.2 All integer Mellin moments can miss a periodic perturbation

The kernel  $\mathcal{W}$  gives a particularly transparent moment-indeterminate family. This is a specific realization of a phenomenon familiar from the classical lognormal moment problem [7], not a claim that moment indeterminacy itself is new.

**Proposition 5.2** (Integer-moment invisibility). *Let  $p : \mathbb{R} \rightarrow \mathbb{R}$  be bounded, measurable, and one-periodic. If*

$$\int_0^\infty \mathcal{W}(t) p(\log_2 t) \frac{dt}{t} = 0, \quad (5.5)$$

*then, for every integer  $k$ ,*

$$\int_0^\infty t^{k-1} \mathcal{W}(t) p(\log_2 t) dt = 0. \quad (5.6)$$

*If in addition  $p$  is nonzero on a set of positive measure and  $\|p\|_\infty \leq 1$ , the probability measures*

$$\frac{\mathcal{W}(t)}{K_0 t} \{1 + \eta p(\log_2 t)\} dt, \quad -1 < \eta < 1, \quad (5.7)$$

*are distinct and have the same moments  $2^{k(k+1)/2}$  for every  $k \in \mathbb{Z}$ .*

*Proof.* Multiplication of  $\mathcal{W}(t)$  by  $p(\log_2 t)$  preserves the scaling equation (3.1). Its Mellin transform therefore satisfies the same shift recurrence as  $K$ , by absolute integrability and the same substitution. A zero at  $s = 0$  consequently gives zeros at all integers. Positivity, normalization, and the moment formula for (5.7) follow immediately. Nontriviality of  $p$  makes the measures distinct.  $\square$

Nonzero functions satisfying (5.5) are easy to produce: subtract the weighted average from any nonconstant bounded periodic function and rescale its sup norm. Later we will impose a second, unweighted mean-zero condition to preserve a gamma normalization as well.

## 6 Classification of completely monotone dyadic solutions

### 6.1 The measure theorem

We use the Bernstein representation theorem: a completely monotone function on  $(0, \infty)$  is the Laplace transform of a unique positive Radon measure on  $[0, \infty)$ , with finite Laplace transform at every positive argument. One reference stating both representation and uniqueness is Aguech–Jedidi [8, Theorem 1].

**Theorem 6.1** (All positive dyadic solutions). *A function  $f : (0, \infty) \rightarrow \mathbb{R}$  is completely monotone and satisfies*

$$f(a) = f(a/2) - f((a+1)/2) \quad (6.1)$$

*if and only if there is a positive Radon measure  $\nu$  on  $(0, \infty)$  with  $\nu(2B) = \nu(B)$  such that*

$$f(a) = \int_0^\infty e^{-at} Q(t) \nu(dt). \quad (6.2)$$

*The measure is unique. It is determined by an arbitrary finite positive measure  $\nu_0$  on the half-open fundamental interval  $[1, 2)$ , and the correspondence is explicitly*

$$f(a) = \int_{[1,2)} \sum_{j \in \mathbb{Z}} e^{-a2^j v} Q(2^j v) \nu_0(dv). \quad (6.3)$$

*The zero measure corresponds to the zero solution.*

*Proof.* Suppose first that  $f$  is completely monotone. Write  $f(a) = \int_{[0,\infty)} e^{-at} \mu(dt)$  by Bernstein's theorem. For a measure  $\mu$ , denote by  $\mu(2 du)$  the pushforward under  $t \mapsto t/2$ , so it assigns mass  $\mu(2B)$  to a Borel set  $B$ . Equation (6.1) and uniqueness of Laplace transforms give

$$\mu(du) = (1 - e^{-u})\mu(2 du). \quad (6.4)$$

This also proves  $\mu(\{0\}) = 0$ . Since  $Q(u) > 0$  for  $u > 0$ , define  $\nu(du) = Q(u)^{-1} \mu(du)$ . It is locally finite on  $(0, \infty)$ . Using  $Q(u) = (1 - e^{-u})Q(2u)$  in (6.4) yields  $\nu(du) = \nu(2 du)$ , exactly the stated invariance. Uniqueness follows from uniqueness of  $\mu$ .

Conversely, let  $\nu$  be positive and invariant. Its mass on  $[1, 2)$  is finite by local finiteness. Partitioning  $(0, \infty)$  into the disjoint half-open intervals  $[2^j, 2^{j+1})$  gives (6.3). At  $j \rightarrow -\infty$ , the bound (2.5) with arbitrarily large  $N$  makes the shell sum converge, even after multiplication by any fixed power of  $t$ . At  $j \rightarrow +\infty$ , the factor  $e^{-at}$  dominates every power. The convergence is locally uniform for  $a > 0$  and persists under all  $a$ -derivatives. Thus (6.2) is finite and completely monotone. The scaling equation of  $Q$  and invariance of  $\nu$  verify (6.1) directly. Finally, a finite measure on  $[1, 2)$  extends uniquely by scaling to such a measure on  $(0, \infty)$ .  $\square$

This classification includes discrete, absolutely continuous, and singular measures. It does not secretly assume that a density with respect to  $dt/t$  exists.

## 6.2 Normalization fixes mass, not phase

**Corollary 6.2** (The normalized solution simplex). *For the solution in Theorem 6.1,*

$$f(1/2) = \nu_0([1, 2)). \quad (6.5)$$

*Consequently, the normalized solutions  $f(1/2) = \lambda$  are exactly the mixtures of*

$$f_\theta(a) = \lambda \sum_{j \in \mathbb{Z}} e^{-a2^{j+\theta}} Q(2^{j+\theta}), \quad 0 \leq \theta < 1. \quad (6.6)$$

*Distinct phases give distinct functions. Their uniform phase average is canonical:*

$$C_0(a) = \int_0^1 f_\theta(a) d\theta. \quad (6.7)$$

*Proof.* The identity

$$e^{-t/2}Q(t) = Q(t) - Q(t/2) \quad (6.8)$$

gives, for every fixed  $v > 0$ ,

$$\sum_{j \in \mathbb{Z}} e^{-2^{j-1}v} Q(2^j v) = \lim_{J \rightarrow \infty} \{Q(2^J v) - Q(2^{-J-1}v)\} = 1.$$

Insert this into (6.3) and use positivity to interchange sum and integral. This proves (6.5). The extremal measures with mass  $\lambda$  are point masses  $\lambda \delta_{2^\theta}$ ; the representation is affine and injective, so their functions are the extremal normalized solutions. Distinct phases have disjoint multiplicative lattices, and uniqueness of Laplace transforms distinguishes the functions. For the last identity, substitute  $t = 2^{j+\theta}$  on each shell, so  $dt/t = \lambda d\theta$ .  $\square$

In particular,

$$C_0(1/2) = \lambda, \quad C_0(1) = \lambda/2, \quad \Gamma_0^{\text{TM}}(1/2) = 2. \quad (6.9)$$

Every normalized solution also has  $f(1) = \lambda/2$ , because the equation at  $a = 1$  reads  $f(1) = f(1/2) - f(1)$ . Thus evaluating only at either of these two points conceals the nonuniqueness. The computations below use  $a = 1/3$  instead.

### 6.3 Nonuniqueness persists for entire differential towers

Given an invariant positive measure  $\nu$ , define, for  $\rho \in \mathbb{R}$ ,

$$C_\rho^\nu(a) = \int_0^\infty t^{-\rho} e^{-at} Q(t) \nu(dt). \quad (6.10)$$

The endpoint argument in the classification proof shows that these integrals are finite and completely monotone for every fixed  $\rho$ . For integer levels,

$$(C_m^\nu)' = -C_{m-1}^\nu \quad (m \geq 1), \quad C_m^\nu(a) = 2^m \{C_m^\nu(a/2) - C_m^\nu((a+1)/2)\}. \quad (6.11)$$

Thus the phase freedom is compatible with every level of the differential ladder and every dyadic equation. Moreover,  $C_m^\nu(a) \rightarrow 0$  as  $a \rightarrow \infty$ , by dominated convergence with any fixed positive lower bound on  $a$ . Vanishing at infinity does not eliminate this freedom.

## 7 Universal coefficients and beyond-all-orders ambiguity

### 7.1 The same coefficient calculus works for every phase measure

For a nonzero invariant positive measure  $\nu$ , set

$$K_\nu(s) = \int_0^\infty t^s \mathcal{W}(t) \nu(dt), \quad \Theta_\nu(\rho) = 2^{-\rho(\rho-1)/2} K_\nu(-\rho). \quad (7.1)$$

The Gaussian envelope on each shell proves that  $K_\nu$  is entire and positive on the real axis. Invariance and (3.1) imply

$$K_\nu(s+1) = 2^{s+1} K_\nu(s). \quad (7.2)$$

Consequently  $\Theta_\nu$  is one-periodic and positive on the real axis.



**Theorem 7.1** (Universal high-order coefficients). *For every nonzero invariant positive measure  $\nu$ , the expansion*

$$\frac{C_\rho^\nu(a)}{2^{\rho(\rho-1)/2}\Theta_\nu(\rho)} = \sum_{k=0}^{N-1} c_k(a)2^{-k\rho} + \mathcal{R}_N^\nu(\rho, a) \quad (7.3)$$

*has exactly the same coefficient polynomials as in Theorem 4.1, and*

$$|\mathcal{R}_N^\nu(\rho, a)| \leq M_{I,r} r^{-N} 2^{N(N+1)/2-N\rho}. \quad (7.4)$$

*In particular, the relative remainder bound is independent of  $\nu$ .*

*Proof.* Write  $\rho = m + \theta$  as before. Scaling by  $2^{-m}$  in the invariant measure and using (4.11) gives

$$C_{m+\theta}^\nu(a) = 2^{m(m-1)/2+m\theta} \int_0^\infty t^{-\theta} \mathcal{W}(t) R_a(2^{-m}t) \nu(dt).$$

Apply the global Taylor estimate and replace the  $K$ -recurrence in the proof of Theorem 4.1 by (7.2). Positivity permits the same absolute remainder estimate. The moment ratios are identical, so no coefficient changes.  $\square$

## 7.2 Normalized towers can agree to every formal order

**Theorem 7.2** (Beyond-all-orders nonuniqueness). *There are infinitely many invariant positive measures  $\nu \neq dt/t$  for which*

$$C_0^\nu(1/2) = \lambda, \quad K_\nu(0) = K_0. \quad (7.5)$$

*For each of them and every fixed  $a > 0$ , the two sequences  $C_m^\nu(a)$  and  $C_m(a)$  have the same complete formal expansion in powers of  $2^{-m}$ , with the same leading constant. More quantitatively, they may be constructed so that, for every compact  $I \subset (0, \infty)$ ,  $0 < r < 4\pi$ , and integer  $N \geq 1$ ,*

$$\frac{|C_m^\nu(a) - C_m(a)|}{K_0 2^{m(m-1)/2}} \leq |\eta| M_{I,r} r^{-N} 2^{N(N+1)/2-Nm}, \quad a \in I, \quad (7.6)$$

*where  $0 < |\eta| < 1$  is a freely chosen perturbation parameter. The corresponding functions  $a \mapsto C_0^\nu(a)$  are nevertheless distinct.*

*Proof.* Choose a nonzero real trigonometric polynomial  $p$  of period one satisfying

$$\int_0^1 p(y) dy = 0, \quad \int_0^\infty \mathcal{W}(t) p(\log_2 t) \frac{dt}{t} = 0, \quad \|p\|_\infty \leq 1. \quad (7.7)$$

Such choices exist in an infinite-dimensional family. To see this concretely, take any finite-dimensional span of nonconstant sines and cosines. Every element already has zero ordinary mean; the weighted condition imposes at most one linear equation. With two or more independent modes there is a nonzero solution, which can be rescaled. Increasing the number of modes gives arbitrarily large families.

Define

$$\nu(dt) = \{1 + \eta p(\log_2 t)\} \frac{dt}{t}, \quad 0 < |\eta| < 1. \quad (7.8)$$

This is positive and dyadically invariant. The first zero mean gives  $\nu([1, 2)) = \lambda$ ; the second gives  $K_\nu(0) = K_0$ . Proposition 5.2 gives equality of all integer  $K$ -moments. Theorem 7.1, at integer orders, therefore gives the same formal expansion and leading constant.

For the stronger bound, use the exact renormalization of the difference:

$$C_m^\nu(a) - C_m(a) = \eta 2^{m(m-1)/2} \int_0^\infty \mathcal{W}(t) p(\log_2 t) R_a(2^{-m}t) \frac{dt}{t}.$$

Every term of the Taylor polynomial integrates to zero by (5.6). The global Taylor remainder,  $|p| \leq 1$ , and  $K(N) = 2^{N(N+1)/2} K_0$  give (7.6). Finally, the Laplace measures  $Q(t)\nu(dt)$  are different because  $p$  is nonzero and  $Q > 0$ . Laplace uniqueness implies that the functions  $C_0^\nu$  are different.  $\square$

Choosing  $N$  as in (4.18)–(4.19), with  $\rho = m$ , makes the relative difference in (7.6) at most

$$|\eta| 2^{1/8} M_{I,r} 2^{-\frac{1}{2}(m+\log_2 r-1/2)^2}. \quad (7.9)$$

This gives a concrete mathematical meaning to “invisible to all formal high-order corrections.” It does not say that two distinct towers have the same exact values. Nor does it rule out characterization by a different asymptotic variable: the next section uses  $a \downarrow 0$  precisely to recover the missing information.

## 8 A boundary characterization of the canonical tower

### 8.1 A single slope limit selects the invariant measure

**Theorem 8.1** (Boundary uniqueness). *Let  $f$  be completely monotone on  $(0, \infty)$  and satisfy (6.1). Then*

$$\lim_{a \downarrow 0} (-a f'(a)) = 1 \quad (8.1)$$

*holds if and only if*

$$f(a) = C_0(a) = \int_0^\infty e^{-at} Q(t) \frac{dt}{t}. \quad (8.2)$$

*No separate point normalization is needed. More generally, replacing the limit one by a nonnegative constant  $c$  selects  $f = cC_0$ .*

*Proof.* Let  $\nu$  be the invariant representing measure from Theorem 6.1. Fix  $s > 0$  and set  $a_N = s2^{-N}$ . Scaling by  $2^N$  in the measure gives

$$-a_N f'(a_N) = \int_0^\infty s t e^{-st} Q(2^N t) \nu(dt). \quad (8.3)$$

For each fixed  $t > 0$ ,  $Q(2^N t) \rightarrow 1$ . The dominating kernel  $s t e^{-st}$  is integrable against  $\nu$ : on shells tending to zero it is  $O_s(2^j)$ , and on shells tending to infinity it has exponential decay. Each shell has the same finite mass. Dominated convergence therefore yields

$$\lim_{N \rightarrow \infty} (-a_N f'(a_N)) = s \int_0^\infty t e^{-st} \nu(dt). \quad (8.4)$$

If (8.1) holds, the right side equals one for every  $s > 0$ . Hence the measure  $t\nu(dt)$  has Laplace transform  $1/s$ . This is the Laplace transform of Lebesgue measure  $dt$  on  $(0, \infty)$ , and uniqueness gives  $t\nu(dt) = dt$ . Thus  $\nu(dt) = dt/t$ , proving necessity.

For the canonical measure, substitution  $u = at$  gives

$$-a C_0'(a) = a \int_0^\infty e^{-at} Q(t) dt = \int_0^\infty e^{-u} Q(u/a) du \rightarrow 1$$

by  $0 < Q \leq 1$  and dominated convergence. The proof with limit  $c$  identifies  $t\nu(dt) = c dt$  in exactly the same way, including  $c = 0$ .  $\square$

Equation (8.4) explains the obstruction in the other solutions: in general, the dyadic boundary limits retain the phase information of  $\nu$ . A full limit as  $a \downarrow 0$ , rather than a limit along one selected dyadic sequence, removes that information. The hypothesis genuinely means a limit through *all* positive  $a$ .

## 8.2 A finite-part condition is sufficient

**Corollary 8.2** (Finite-part characterization). *Within the completely monotone solutions of (6.1), the existence of a finite limit*

$$\lim_{a \downarrow 0} \{f(a) + \log a\} \in \mathbb{R} \quad (8.5)$$

*is equivalent to  $f = C_0$ . For the canonical function, the limit is*

$$\kappa = -\gamma + \int_0^1 \frac{Q(t)}{t} dt + \int_1^\infty \frac{Q(t) - 1}{t} dt, \quad (8.6)$$

*where  $\gamma$  is Euler's constant.*

*Proof.* Complete monotonicity implies convexity. For  $0 < h < 1$ , the two secant bounds around  $a$  give

$$\frac{f(a) - f((1+h)a)}{h} \leq -af'(a) \leq \frac{f((1-h)a) - f(a)}{h}. \quad (8.7)$$

Under (8.5), the left and right expressions tend, as  $a \downarrow 0$ , to  $\log(1+h)/h$  and  $-\log(1-h)/h$  respectively. Letting  $h \downarrow 0$  proves (8.1), and Theorem 8.1 applies.

For the converse, split the canonical integral as

$$C_0(a) = \int_0^1 e^{-at} \frac{Q(t)}{t} dt + \int_1^\infty e^{-at} \frac{Q(t) - 1}{t} dt + \int_1^\infty e^{-at} \frac{dt}{t}.$$

The first two integrals converge to their values at  $a = 0$  by Lemma 2.1. The last is the exponential integral  $E_1(a) = -\log a - \gamma + o(1)$ , which gives (8.6).  $\square$

The finite-part hypothesis is stronger than the slope hypothesis as a condition on arbitrary functions, but both select exactly the same member of the dyadic completely monotone family.

## 8.3 Uniqueness of the full differential hierarchy

**Corollary 8.3** (Canonical gamma-tower characterization). *Suppose  $f_0, f_1, \dots$  are completely monotone functions on  $(0, \infty)$  satisfying*

$$\begin{aligned} f_m(a) &= 2^m \{f_m(a/2) - f_m((a+1)/2)\} \quad (m \geq 0), \\ f'_m(a) &= -f_{m-1}(a) \quad (m \geq 1), \end{aligned} \quad (8.8)$$

*and  $-af'_0(a) \rightarrow 1$  as  $a \downarrow 0$ . Then  $f_m = C_m$  for every  $m \geq 0$ . Equivalently, the functions  $\exp((-1)^m m! f_m)$  are exactly the hierarchy in (4.20).*

*Proof.* Theorem 8.1 gives  $f_0 = C_0$ . Inductively, if  $f_{m-1} = C_{m-1}$ , the differential ladder says that  $f_m - C_m$  is a constant  $d_m$ . Subtracting the two level- $m$  dyadic equations gives  $d_m = 2^m(d_m - d_m) = 0$ . This completes the induction.  $\square$

This supplies a normalization and a proof of uniqueness in a precise positive class. It does not establish every possible Weierstrass-product or complex-analytic assertion one might include in a broader Barnes-type program. The two issues should be kept separate.

## 9 Reproducible computations and what they verify

### 9.1 Independent integral evaluation

The archive contains `verify_results.py` and two JSON reports. The coefficient identities were checked exactly with rational symbolic arithmetic through  $k = 12$ . The numerical calculations were performed with 48 Gauss–Legendre nodes at 60 decimal digits and repeated with 80 nodes at 80 digits. Both the precision and the quadrature size were increased; agreement is not based solely on changing the displayed number of digits.

For  $K_0$  and  $\Theta$ , the program evaluates a single multiplicative period. Combining (3.3) with  $t = 2^{y+j}$  gives

$$\Theta(\rho) = \lambda 2^{1/8} \int_0^1 A(y) \sum_{j \in \mathbb{Z}} 2^{-(j+y+\rho-1/2)^2/2} dy. \quad (9.1)$$

Periodicity permits reduction of  $\rho$  modulo one. The rapidly convergent Gaussian shell sum makes this inexpensive.

The  $C_\rho$  checks are evaluated separately from their *original*  $Q$ -integrals, not by inserting the claimed coefficient expansion or the  $R_a$ -renormalization. On logarithmic shells the computation uses

$$\begin{aligned} \frac{C_\rho(a)}{2^{\rho(\rho-1)/2}} &= \lambda \int_0^1 \sum_{j \in \mathbb{Z}} \exp(-\rho\lambda(y+j) - a2^{y+j}) \\ &\quad + \log Q(2^{y+j}) - \frac{\lambda}{2}\rho(\rho-1)) dy. \end{aligned} \quad (9.2)$$

Small factors are evaluated with `expm1` to avoid cancellation. The shell values of  $Q$  use its own product recurrence. This comparison can detect an incorrect prefactor or a wrong coefficient in the theorem.

The two runs agree far beyond the digits printed below. For example, the reported  $K_0$  values agree in the first 49 decimal places; their stored difference is less than  $4 \cdot 10^{-51}$ . The calculations are *not interval-certified*: quadrature stability and exact symbolic checks support the implementation, while the proofs establish the theorems. The numerical work does not establish a sharp constant in the remainder bound or historical novelty.

### 9.2 Centered derivative data

Let

$$Z_m = \frac{C_m(1/2)}{K_0 2^{m(m-1)/2}}, \quad P_2 = 1 - \frac{2^{-2m}}{9}, \quad P_4 = P_2 + \frac{248}{2025} 2^{-4m}, \quad P_6 = P_4 - \frac{4806656}{2679075} 2^{-6m}.$$

| $m$ | $ Z_m - 1 $               | $ Z_m - P_2 $             | $ Z_m - P_4 $             | $ Z_m - P_6 $             |
|-----|---------------------------|---------------------------|---------------------------|---------------------------|
| 4   | $4.322382 \cdot 10^{-4}$  | $1.789613 \cdot 10^{-6}$  | $7.911730 \cdot 10^{-8}$  | $2.782223 \cdot 10^{-8}$  |
| 8   | $1.695392 \cdot 10^{-6}$  | $2.850821 \cdot 10^{-11}$ | $6.353778 \cdot 10^{-15}$ | $2.031475 \cdot 10^{-17}$ |
| 12  | $6.622738 \cdot 10^{-9}$  | $4.350974 \cdot 10^{-16}$ | $3.799206 \cdot 10^{-22}$ | $4.918108 \cdot 10^{-27}$ |
| 16  | $2.587007 \cdot 10^{-11}$ | $6.639065 \cdot 10^{-21}$ | $2.264533 \cdot 10^{-29}$ | $1.145315 \cdot 10^{-36}$ |

Table 1: Absolute errors in the normalized centered expansion. The notation  $x \cdot 10^{-j}$  means  $x \times 10^{-j}$ .

Table 1 compares successive approximations. For fixed truncation order the improvement agrees with the even powers predicted by symmetry.

At  $m = 4$ , the benefit of further terms is already much less spectacular than at  $m = 16$ . This is consistent with the fact that the expansion is divergent: adding indefinitely many terms at a fixed order is not justified. The full data files also include nonsymmetric shifts and noninteger orders, for example  $a = 1$  and  $\rho = 8.25, 8.5, 12.75$ .

### 9.3 The tiny Mellin phase and the visible gamma phase

The observed relative modulation of  $\Theta$  is shown in Table 2. Its small size is consistent with the exact Fourier smoothing factor in (3.9).

| $\theta$ | $\Theta(\theta)/\Theta(0) - 1$         |
|----------|----------------------------------------|
| 1/4      | $1.289029564918397699 \times 10^{-18}$ |
| 1/2      | $1.296624062281340296 \times 10^{-18}$ |
| 3/4      | $7.594497362942597195 \times 10^{-21}$ |

Table 2: Numerical phase variation. Nonconstancy is numerical evidence here, not an additional theorem assumed elsewhere.

For comparison, Table 3 gives the explicit extremal solutions of (6.6). Their differences are much easier to see, although all have  $f_\theta(1/2) = \lambda$  and  $f_\theta(1) = \lambda/2$ .

| $\theta$ | $f_\theta(1/3)$      | $-af'_\theta(a)$ at $a = 2^{-24}$ |
|----------|----------------------|-----------------------------------|
| 0        | 0.956669839913615728 | 1.00000628193280958               |
| 1/4      | 0.956669741884495862 | 0.99999235648832259               |
| 1/2      | 0.956665362383835818 | 0.99999357544426848               |
| 3/4      | 0.956665460410626991 | 1.00000750091845969               |

Table 3: Distinct positive normalized solutions. The canonical value is  $C_0(1/3) = 0.956667601148143600\dots$ . The slope column is a finite- $a$  evaluation, not a claimed exact limiting value.

The program also checks the dyadic equation away from these special normalization points, using  $a = 1.3$  and  $\theta = 0.27$ . The residual is below  $10^{-70}$  in the 80-digit run. Agreement at  $a = 1$  alone would be an inadequate test because that value is forced by the normalization.

## 9.4 How to reproduce the data

From the extracted archive, run

```
python verify_results.py --dps 60 --nodes 48 --output results_48.json
python verify_results.py --dps 80 --nodes 80 --output results_80.json
```

The script requires `mpmath` and `sympy`, records their installed versions, and has no network dependency. It checks the coefficient derivative law, reflection law, and centered parity by exact symbolic simplification. It does not use numerical differentiation of  $D$  near its zeros, where cancellation would make a direct approach much less stable.

## 10 Interpretation and further research directions

The principal mechanism is not cancellation among many alternating Dirichlet terms. It is a positive product identity. Once  $Q$  is paired with its dyadic uniform-sum counterpart  $H$ , the resulting multiplicative cocycle makes its Mellin moments exactly solvable. The difficult high-order dependence moves into a single analytic function  $R_a$  at a small argument. This is why explicit coefficients and a global remainder can be obtained together.

The classification theorem gives a complementary lesson. The dyadic functional equation controls how a measure moves between scales, but it does not prescribe the distribution of mass within a scale. One-point normalization determines only that scale’s total mass. The all-orders construction shows that even a full formal expansion at integer orders can fail to recover the missing phase information. The boundary slope recovers the whole Laplace transform of the invariant measure, and thus is strong enough to characterize it.

Several concrete extensions remain worth pursuing. One is to obtain a closed Fourier or residue formula for the amplitude  $A$ , and from it a fully explicit proof and bound for the nonconstant modulation of  $\Theta$ . Another is to determine the leading exponentially small discrepancy in Theorem 7.2, rather than only its Gaussian-in-order upper bound. The poles of  $R_a$  at imaginary multiples of  $4\pi$  and the chosen phase measure should enter such a refinement, but this article does not supply that additional expansion.

A separate quantitative task is to replace the tube supremum  $M_{I,r}$  by simple computable constants optimized for a given shift interval. The proof of Lemma 4.3 already separates a compact region from a controlled tail, so interval arithmetic could turn it into a numerical certificate. The present tables should not be mistaken for that certificate.

The positive-measure classification and the proof of the uniform remainder also have clear formalization targets: product convergence, pushforward identities, the Mellin recurrence, Cauchy estimates on a tube, and Laplace uniqueness. None of these targets requires asserting that the results already appear in the repository’s Lean code. Finally, analogous product pairs for other dilation factors may produce related invariant-measure classifications; a radix change must be made at the level of the product equation rather than by replacing every two in the final formulas without checking the underlying sequence.

## A A compact audit of hypotheses and conclusions

The asymptotic parameter is real  $\rho \rightarrow \infty$ , with  $a$  in a fixed compact subset of  $(0, \infty)$ . Uniformity up to  $a = 0$  is not claimed: indeed,  $R_0(x) = 1/H(x)$  is unbounded as  $x \rightarrow \infty$ , which invalidates the tube supremum used in the proof. The restriction is structural rather than cosmetic.

The all-orders series has zero convergence radius as a series in  $h = 2^{-\rho}$ . A finite expansion with a proved remainder is the intended meaning of every  $\sim$  symbol in this article. Optimal truncation gives an upper bound of Gaussian size in  $\rho$ , but not a matching lower bound. The finite coefficient formula is exact independently of numerical tests.

The uniqueness theorem assumes complete monotonicity, not merely positivity of  $f$  or of an exponentiated gamma function. This assumption makes a positive Laplace measure available and is essential to the stated classification proof. The normalized nonunique examples satisfy this stronger condition, so they are not counterexamples obtained by leaving the proposed positive class.

The ambiguity theorem compares integer-order expansions. In general its perturbations need not preserve the entire function  $\Theta_\nu(\rho)$  at noninteger orders. Equality of every integer Mellin moment is weaker than equality of the full Mellin transform, just as the explicit Stieltjes family demonstrates.

The term “gamma tower” refers exactly to  $\exp((-1)^m m! C_m(a))$ , or its measure-deformed counterpart. The article proves a real positive-class characterization; it does not silently assume a preferred continuation to all complex  $a$ , a canonical Weierstrass factorization, or an inversion transseries for that tower.

## B Source and novelty audit

The supplied repository document was read as the public raw LaTeX source at the URL in reference [1]; its consolidated document is dated August 2026. The comparison used its Mellin-continuation material, its automatic gamma-hierarchy material, and its stated frontier on hierarchy normalization. A moving repository URL is not an immutable version pin; no unverified commit hash is assigned to it here.

Classical Dirichlet-series facts were checked against the sources listed below. The primary text of the complete-monotonicity representation was available in [8]; the present measure equation and its classification are derived in Section 6, rather than attributed to that reference. The general historical statements about Mellin log-periodicity and lognormal moment indeterminacy are background only. None of those references is being credited with the particular uniform remainder theorem or the boundary argument without evidence.

The 2001 Alkauskas preprint was located on its author’s bibliography, but the linked PDF did not load during this investigation. It is recorded because of its potentially relevant subject, not cited as a text that was fully examined. Consequently, historical priority remains a separate verification task. The claims of this article are the stated mathematical theorems and their proofs, supplemented by reproducible but non-interval numerical checks.

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